SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – Data Interpretation

Course Code:	DSA03	**Course Title:**	Natural Language Processing
CO Assessed:	CO4 – Precision	**Assessment:**	Assessment Tool 1 – Data Interpretation
Weightage:	40% of CIA		
CO5 Statement:	Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4)		
Student Name:	Shaik Nuha	**Reg. No.:**	192472242
Section / Batch:	B.E-CSE-AI	**Date:**	

Code of Conduct

I __________Shaik Nuha(19272242)______ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: __________Sk.Nuha_____________

Instructions

- Read each data set, table, semantic representation, or scenario carefully before answering.
- Analyze the given information and provide logical, evidence-based interpretations.
- Show all intermediate reasoning, semantic analysis steps, predicate representations, or calculations wherever applicable.
- Answers should be concise, structured, and technically accurate.
- Support your conclusions using the data provided in the question.
- Draw diagrams, semantic networks, parse trees, or logical representations wherever relevant.
- Avoid copying definitions alone; focus on analyzing the given scenario and interpreting the data.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Representing Meaning & Meaning Structure of Language	Analyze semantic representations, identify action–entity relationships, detect interpretation errors, and evaluate meaning representation in chatbot systems.	Q1
First-Order Predicate Calculus & Representing Linguistically Relevant Concepts	Construct predicate representations, apply logical inference rules, derive conclusions, and evaluate reasoning in smart manufacturing environments.	Q2
Lexemes and Their Senses & Word Sense Disambiguation	Analyze contextual clues, determine correct word senses, evaluate ambiguity resolution, and improve semantic understanding in search systems.	Q3
Syntax-Driven Semantic Analysis	Analyze syntactic structures, assign semantic roles, evaluate parsing accuracy, and improve semantic interpretation in healthcare NLP applications.	Q4

Annexure A – Data Interpretation

1. Semantic Representation in Customer Support Chatbots.
   A telecom company collected customer chatbot interactions. The semantic analyzer generated the following meaning representations.

Customer Query	Semantic Representation
"Activate international roaming for my number."	ACTIVATE(Roaming, Customer)
"Deactivate caller tune service."	DEACTIVATE(CallerTune, Customer)
"Check my data balance."	QUERY(DataBalance, Customer)
"Enable 5G service."	ACTIVATE(5GService, Customer)

Additional chatbot logs show:

Query ID	Actual User Intent	Predicted Intent
Q1	Activate Roaming	Activate Roaming
Q2	Deactivate Caller Tune	Activate Caller Tune
Q3	Check Data Balance	Query Data Balance
Q4	Enable 5G Service	Activate 5G Service

Tasks:

1. Analyze the semantic representations and identify the action-object relationships.
2. Determine which query contains semantic interpretation errors.
3. Evaluate how meaning representation affects chatbot decision accuracy.
4. Recommend improvements to enhance semantic understanding in telecom customer support systems.

2. First-Order Predicate Calculus for Smart Manufacturing.
   An Industry 4.0 manufacturing plant monitors machines using semantic rules.
   Production Data:

Machine	Status
M1	Active
M2	Active
M3	Maintenance
M4	Active

Predicate Rules:

- Active(x) → Producing(x)
- Produces(x,y) ∧ Active(x) → Available(y)
- Maintenance(x) → ¬Producing(x)

Tasks:

1. Represent the production data using First-Order Predicate Calculus.
2. Analyze which products are currently available.
3. Infer whether Gear production is affected by maintenance activities.
4. Evaluate the effectiveness of predicate logic in industrial decision-support systems.

3. Word Sense Disambiguation in E-Commerce Search Engines.
   An e-commerce company observed the following search logs.

User Query	Possible Sense 1	Possible Sense 2
"Apple accessories"	Fruit	Technology Brand
"Mouse wireless"	Animal	Computer Device
"Java tutorial"	Island	Programming Language
"Python course"	Snake	Programming Language

Search Context Data:

Query	Clicked Result
Apple accessories	iPhone Charger
Mouse wireless	Bluetooth Mouse
Java tutorial	Coding Lessons
Python course	Software Development Training

Tasks:

1. Analyze the contextual information and determine the correct word sense for each query.
2. Identify the semantic cues used for disambiguation.
3. Evaluate the impact of incorrect sense selection on search engine performance.
4. Suggest an industrial-scale WSD strategy for improving recommendation accuracy.

4. Syntax-Driven Semantic Analysis in Healthcare Information Systems.

A healthcare NLP system processes medical reports.

Parsed Sentences:

Sentence	Parse Structure
Doctor prescribed medicine to patient.	Subject-Verb-Object
Patient reported severe headache.	Subject-Verb-Object
Nurse monitored patient continuously.	Subject-Verb-Object
Medicine reduced blood pressure.	Subject-Verb-Object

Semantic Roles Generated:

Entity	Role
Doctor	Agent
Medicine	Instrument
Patient	Recipient
Headache	Symptom

Tasks:

1. Analyze how syntactic structures contribute to semantic role assignment.
2. Determine whether the assigned semantic roles are appropriate.
3. Evaluate potential errors that could arise from incorrect parsing.
4. Recommend methods to improve syntax-driven semantic analysis in healthcare applications.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

Q. No. / Criteria Level	Understanding of Data	Analysis	Application of Concepts	Conclusion	Clarity	Sub. Total	Total %
1							
2							
3							
4							

Faculty In-charge	Course Coordinator	Program Director
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________

1. Semantic Representation in Customer Support Chatbots

Task 1: Analyse the semantic representations and identify the action-object relationships

Semantic representation converts a natural-language customer query into a structured meaning representation. In the given examples, the general form is:

ACTION (Object, Customer)

Customer Query	Semantic Representation	Action	Object
Activate international roaming for my number.	ACTIVATE (Roaming, Customer)	ACTIVATE	Roaming
Deactivate caller tune service.	DEACTIVATE (CallerTune, Customer)	DEACTIVATE	Caller Tune
Check my data balance.	QUERY (DataBalance, Customer)	QUERY	Data Balance
Enable 5G service.	ACTIVATE (5GService, Customer)	ACTIVATE	5G Service

The semantic representation correctly captures the intent/action and the target service/object of each query.

For example:

ACTIVATE (Roaming, Customer)

means that the Customer wants the Roaming service to be activated.

Task 2: Determine which query contains semantic interpretation errors

The incorrect interpretation occurs in Q2.

Query ID	Actual Intent	Predicted Intent	Result
Q1	Activate Roaming	Activate Roaming	Correct
Q2	Deactivate Caller Tune	Activate Caller Tune	**Incorrect**
Q3	Check Data Balance	Query Data Balance	Correct
Q4	Enable 5G Service	Activate 5G Service	Correct

Error in Q2

Actual:

DEACTIVATE (CallerTune, Customer)

Predicted:

ACTIVATE (CallerTune, Customer)

The object CallerTune is correctly identified, but the action is incorrect.

The system has confused:

DEACTIVATE → ACTIVATE

This is a significant semantic error because it can cause the chatbot to perform the opposite operation requested by the customer.

Task 3: Evaluate how meaning representation affects chatbot decision accuracy

Meaning representation directly affects chatbot decision-making.

A chatbot generally follows this process:

User Query → Semantic Analysis → Intent Detection → Action Selection → System Response

If semantic representation is correct, the chatbot can perform the appropriate operation.

For example:

ACTIVATE (Roaming, Customer)

can trigger:

Activate International Roaming

However, if the representation is incorrect:

ACTIVATE (CallerTune, Customer)

instead of:

DEACTIVATE (CallerTune, Customer)

the chatbot may activate a service that the customer actually wants to deactivate.

Therefore, accurate semantic representation improves:

- Intent classification
- Service activation/deactivation
- Customer satisfaction
- Automated decision-making
- Response accuracy
- Reduction of incorrect transactions

Task 4: Recommendations to improve semantic understanding

The telecom chatbot can be improved using the following techniques:

1. **Intent classification**
   Train models to distinguish between actions such as activate, deactivate, query, renew, and cancel.
2. **Entity recognition**
   Identify telecom entities such as:
   - Roaming
   - 5G
   - Data Balance
   - Caller Tune
3. **Context-aware analysis**
   Consider previous conversation turns when interpreting ambiguous queries.
4. **Negation detection**
   Explicitly detect words such as:
   - not

- deactivate
- cancel
- stop
- disable

5. **Ontology-based representation**
   Maintain a telecom ontology containing services, actions, and relationships.
6. **Confidence scoring**
   If confidence is low, the chatbot should ask for clarification instead of performing an incorrect operation.

Conclusion

Semantic representation provides the bridge between natural language and chatbot actions. Correct identification of both action and object is essential for reliable telecom customer support.

2. First-Order Predicate Calculus for Smart Manufacturing

Task 1: Represent the production data using First-Order Predicate Calculus

The production data can be represented using predicates.

Machine Status Facts

Given:

- M1 is Active
- M2 is Active
- M3 is under Maintenance
- M4 is Active

The corresponding predicates are:

Active(M1)

Active(M2)

Maintenance(M3)

Active(M4)

Predicate Rules

The given rules are:

Active(x) → Producing(x)

Produces (x, y) ∧ Active(x) → Available(y)

Maintenance(x) → ¬Producing(x)

These rules describe relationships between machine status, production, and product availability.

Task 2: Analyse which products are currently available

From:

Active(M1)

Active(M2)

Active(M4)
and:
Active(x) → Producing(x)
we can infer:
Producing(M1)
Producing(M2)
Producing(M4)
For M3:
Maintenance(M3)
Using:
Maintenance(x) → ¬Producing(x)
we infer:
¬Producing(M3)
However, the given production data does not explicitly specify which machine produces which product.
The rule:
Produces (x, y) ∧ Active(x) → Available(y)
requires a fact such as:
Produces (M1, Gear)
before we can infer:
Available (Gear)
Therefore:
The available products cannot be uniquely determined from the supplied facts alone.
For example, if the plant additionally contains:
Produces (M1, Gear)
Produces (M3, Gear)
then:
Available (Gear)
because M1 is active, even though M3 is under maintenance.

Task 3: Infer whether Gear production is affected by maintenance activities
Suppose:
Produces (M3, Gear)
and:
Maintenance(M3)
Then:
Maintenance(M3) → ¬Producing(M3)
Therefore, M3 cannot currently produce Gear.
However, if another active machine also produces Gear:
Produces (M1, Gear)

and:
Active(M1)
then:
Available (Gear)
can still be inferred.
Thus, maintenance affects the production capacity of the particular machine, but it does not necessarily mean that the product becomes completely unavailable.

Task 4: Evaluate the effectiveness of predicate logic
First-Order Predicate Calculus is useful in industrial decision-support systems because it provides:

- Formal representation of machine states
- Logical inference
- Rule-based decision-making
- Clear relationships between machines and products
- Explainable conclusions
- Easy integration of additional rules

For example:
Maintenance(M3)
combined with:
Maintenance(x) → ¬Producing(x)
allows the system to automatically infer:
¬Producing(M3)
Limitation:
Predicate logic requires sufficiently complete knowledge. In this problem, product availability cannot be determined without Produces (machine, product) facts.
Conclusion:
Predicate logic is highly effective for rule-based industrial reasoning, particularly when machine states and production relationships are explicitly represented.

3. Word Sense Disambiguation in E-Commerce Search Engines
Task 1: Determine the correct word sense using contextual information
The correct senses are:

User Query	**Possible Senses**	**Context / Clicked Result**	**Correct Sense**
Apple accessories	Fruit / Technology Brand	iPhone Charger	Technology Brand
Mouse wireless	Animal / Computer Device	Bluetooth Mouse	Computer Device

Java tutorial	Island / Programming Language	Coding Lessons	Programming Language
Python course	Snake / Programming Language	Software Development Training	Programming Language

Analysis:

Apple accessories

The clicked result is iPhone Charger, which clearly indicates the technology company/product ecosystem rather than the fruit.

Mouse wireless

The result Bluetooth Mouse indicates a computer peripheral.

Java tutorial

The result Coding Lessons indicates the Java programming language.

Python course

The result Software Development Training indicates Python as a programming language.

Task 2: Identify semantic cues used for disambiguation

The main semantic cues are:

1. Query context

Words surrounding the ambiguous term provide clues.

For example:

Java + tutorial

suggests programming rather than an island.

2. Clicked result

The clicked result provides strong evidence about the user's intended meaning.

Examples:

iPhone Charger → Apple = Technology

Bluetooth Mouse → Mouse = Computer Device

Coding Lessons → Java = Programming

3. Domain-specific vocabulary

Words such as:

- accessories
- tutorial
- course
- coding
- software
- Bluetooth

help identify the correct sense.

4. User behaviour

Click history and interaction data can help determine user intent.

Task 3: Impact of incorrect sense selection

Incorrect WSD can significantly reduce search-engine performance.

For example, if:

"Python course" is interpreted as the snake, the system may return:

- Snake biology
- Python species
- Wildlife information

instead of:

- Python programming courses
- Coding tutorials
- Software development training

This can result in:

- Low search relevance
- Poor recommendations
- Lower click-through rate
- Poor user experience
- Incorrect personalization
- Reduced customer engagement

Therefore, accurate WSD is critical for e-commerce search and recommendation systems.

Task 4: Industrial-scale WSD strategy

A large e-commerce company can use a hybrid WSD architecture:

Step 1: Query preprocessing

Perform:

- Tokenization
- Normalization
- Entity recognition

Step 2: Context analysis

Use:

- Query terms
- Previous queries
- Search history
- Click history
- Product categories

Step 3: Candidate sense generation

Generate possible meanings for ambiguous words.

Step 4: Contextual language model

Use transformer-based models to calculate which sense best fits the context.

Step 5: Behavioural signals

Use:

- Click-through rate
- Purchase history
- Dwell time
- Product interactions

Step 6: Ranking

Rank candidate senses and select the highest-confidence interpretation.

Step 7: Feedback

Use user interactions as feedback to continuously improve the WSD model.

Conclusion:

A combination of contextual NLP, knowledge bases, behavioral signals, and machine-learning models provides more accurate word-sense disambiguation at industrial scale.

4. Syntax-Driven Semantic Analysis in Healthcare Information Systems

Task 1: Analyse how syntactic structures contribute to semantic role assignment

Syntactic parsing identifies relationships between words in a sentence. These relationships help determine semantic roles such as Agent, Patient, Instrument, Recipient, and Symptom.

For example:

Sentence: Doctor prescribed medicine to patient.

Syntactic structure:

Subject → Doctor

Verb → prescribed

Object → medicine

Indirect Object → patient

Semantic interpretation:

Doctor → Agent

Medicine → Theme

Patient → Recipient

Similarly:

Patient reported severe headache.

Patient → Subject → Experiencer

Headache → Object → Symptom

Thus, syntax provides important structural information for semantic role labelling.

Task 2: Determine whether the assigned semantic roles are appropriate

The given roles are partly correct but require some correction.

Entity	Given Role	Evaluation	Better Role
Doctor	Agent	Correct	Agent

Medicine	Instrument	Not ideal	Theme
Patient	Recipient	Correct in first sentence	Recipient
Headache	Symptom	Correct	Symptom

Important correction

In: Doctor prescribed medicine to patient.

The **doctor** is the Agent because the doctor performs the prescribing action.

The **patient** is the Recipient.

The **medicine** is generally the **Theme**, because it is the entity being prescribed/transferred.

Calling medicine an **Instrument** is not appropriate in this sentence.

An instrument is normally something used to perform an action, such as:

Doctor examined patient with a stethoscope.

Here:

Stethoscope → Instrument

Task 3: Potential errors from incorrect parsing

Incorrect syntactic parsing can cause serious errors in healthcare NLP.

Example

Correct:

Doctor prescribed medicine to patient.

If the parser incorrectly identifies patient as the Agent, the system could interpret the sentence as:

Patient prescribed medicine.

This completely changes the medical meaning.

Potential consequences include:

- Incorrect patient records
- Wrong medication relationships
- Incorrect clinical summaries
- Incorrect medical question answering
- Faulty decision-support recommendations
- Misinterpretation of symptoms
- Incorrect extraction of treatment information

Healthcare systems therefore require highly accurate syntax and semantic analysis.

Task 4: Recommendations to improve syntax-driven semantic analysis

1. Domain-specific NLP models

Use models trained specifically on medical text.

2. Dependency parsing

Use dependency parsing to identify grammatical relationships between medical entities.

3. Semantic Role Labelling

Apply SRL to identify:

- Agent
- Patient
- Theme
- Recipient
- Instrument
- Experiencer

4. Medical ontologies

Integrate resources such as medical terminology databases and ontologies.

5. Named Entity Recognition

Identify:

- Diseases
- Symptoms
- Medicines
- Doctors
- Patients
- Procedures

6. Contextual language models

Use transformer-based models capable of understanding context and medical terminology.

7. Human validation

For high-risk clinical applications, uncertain interpretations should be reviewed by medical professionals.

Conclusion:

Syntax-driven semantic analysis is important for converting medical text into structured clinical information. Combining syntactic parsing, semantic role labelling, medical NER, domain knowledge, and contextual NLP models can significantly improve accuracy and reliability.